

VPA ICT — IGCSE Y9S1 Weeks 1–9: Computer Systems & Networks

Weeks 1 – 9: Computer Systems, Input/Output, Storage & Networks

Syllabus: CIE IGCSE ICT (0417) Topics 1–4 | Lesson Length: 40 minutes

Week 1 — Types of Computer Systems

Syllabus: 1.1 Types and Components of Computer Systems

Keywords: mainframe 大型机 (dàxíng jī), supercomputer 超级计算机 (chāojí jìsuànjī), personal computer 个人电脑 (gèrén diànnǎo), laptop 笔记本电脑 (bǐjìběn diànnǎo), tablet 平板电脑 (píngbǎn diànnǎo), smartphone 智能手机 (zhìnéng shǒujī), embedded system 嵌入式系统 (qiànrù shì xítǒng)

Activity 1 (15 min): Matching game — Match computer type to its typical use case (hospital, weather forecasting, home, car).

Activity 2 (15 min): Comparison table — Compare mainframe, supercomputer, PC, embedded on: speed, cost, size, typical use.

Activity 3 (10 min): Discussion — "Is a smartwatch a computer?" Debate using definition criteria.

Exam Tip: For "compare" questions, always give a balanced answer. Don't just describe one type — explicitly state similarities AND differences.

Week 2 — Hardware Components

Syllabus: 1.2 Hardware — CPU, Memory, Motherboard

Keywords: CPU 中央处理器 (zhōngyāng chǔlǐ qì), ALU 算术逻辑单元 (suànsù luójí dānyuán), CU 控制单元 (kòngzhì dānyuán), register 寄存器 (jìcúqì), RAM 随机存取存储器 (suíjī cúnqǔ cúnchǔqì), ROM 只读存储器 (zhǐdú cúnchǔqì), motherboard 主板 (zhǔbǎn), bus 总线 (zǒngxiàn)

Activity 1 (15 min): CPU diagram labeling — Students label ALU, CU, registers, cache on a diagram.

Activity 2 (15 min): RAM vs. ROM comparison — Create a Venn diagram. What happens when power is lost?

Activity 3 (10 min): "Fetch-Decode-Execute" cycle — Students act out the cycle in groups of 3.

Exam Tip: The fetch-decode-execute cycle is a common 4-mark question. State: 1) Address sent to memory, 2) Instruction fetched, 3) Instruction decoded by CU, 4) Instruction executed by ALU.

Week 3 — Input Devices

Syllabus: 2.1 Input Devices

Keywords: keyboard 键盘 (jiàn pán), mouse 鼠标 (shǔ biāo), scanner 扫描仪 (sǎo miáo yí), microphone 麦克风 (mài kè fēng), webcam 网络摄像头 (wǎng luò shè xiàng tóu), barcode reader 条码阅读器 (tiáo mǎ yuè dú qì), RFID 射频识别 (shè pín shí bié), sensor 传感器 (chuángǎn qì), OCR 光学字符识别 (guāng xué zì fú shí bié), OMR 光学标记识别 (guāng xué biāo jì shí bié)

Activity 1 (15 min): Input device scavenger hunt — Find 5 input devices in the lab; photograph and label.

Activity 2 (15 min): Scenario matching — Given situations (supermarket, hospital, factory), choose the most appropriate input device and justify.

Activity 3 (10 min): OCR demonstration — Scan a printed page; compare accuracy to typed text.

Exam Tip: "Suggest a suitable input device for..." questions require justification. Always state: device name + why it's suitable + why alternatives are less suitable.

Week 4 — Output Devices

Syllabus: 2.2 Output Devices

Keywords: monitor 显示器 (xiǎnshìqì), printer 打印机 (dǎyìnjī), inkjet 喷墨打印机 (pēnmò dǎyìnjī), laser 激光打印机 (jīguāng dǎyìnjī), 3D printer 3D打印机 (3D dǎyìnjī), speaker 扬声器 (yángshēngqì), actuator 执行器 (zhíxíngqì), plotter 绘图仪 (huìtú yí), projector 投影仪 (tóuyǐng yí)

Activity 1 (15 min): Printer comparison — Compare inkjet vs. laser on: cost per page, speed, quality, use case.

Activity 2 (15 min): Output device role-play — One student is an architect, one a gamer, one a warehouse manager. Others suggest and justify output devices.

Activity 3 (10 min): "Output for disabilities" — Research and present: Braille printer, screen reader, haptic feedback.

Week 5 — Storage Devices & Media

Syllabus: 3.1–3.2 Storage Devices and Media

Keywords: hard disk drive (HDD) 硬盘驱动器 (yìngpán qūdòngqì), solid state drive (SSD) 固态硬盘 (gùtài yìngpán), USB flash drive U盘 (U pán), optical disc 光盘 (guāngpán), magnetic tape 磁带 (cídài), cloud storage 云存储 (yún cúnchǔ), NAS 网络附加存储 (wǎngluò fùjiā cúnchǔ), read/write speed 读写速度 (dúxiě sùdù), capacity 容量 (róngliàng), portability 便携性 (biànxíe xìng)

Activity 1 (15 min): Storage comparison matrix — Evaluate HDD, SSD, USB, DVD, Cloud on: capacity, speed, cost, portability, durability.

Activity 2 (15 min): "Best storage for..." — Scenarios: video editor, traveler, school archive, gamer.

Activity 3 (10 min): Calculate storage needs — How much space for 1000 photos (5MB each)? 10 hours of video (1GB/hour)?

Exam Tip: For storage comparison questions, use the CAPS framework: Capacity, Access speed, Portability, Suitability (durability/cost).

Week 6 — Data Storage Units & Compression

Syllabus: 3.3 Data Storage — Bits, Bytes, and Compression

Keywords: bit 位 (wèi), byte 字节 (zìjié), kilobyte 千字节 (qiān zìjié), megabyte 兆字节 (zhào zìjié), gigabyte 吉字节 (jí zìjié), terabyte 太字节 (tài zìjié), compression 压缩 (yāsuō), lossy 有损压缩 (yǒusǔn yāsuō), lossless 无损压缩 (wúsuǎn yāsuō)

Activity 1 (15 min): Conversion practice — Convert between bits, bytes, KB, MB, GB. Mental math: How many bytes in 2KB?

Activity 2 (15 min): Compression experiment — Save same image as BMP, PNG, JPG. Compare file sizes. Zoom in — what changed?

Activity 3 (10 min): Calculate image file size: 1024×768 pixels, 24-bit color = ? MB

File size = width $\times$ height $\times$ color depth (bits) $\div 8$ (to bytes) $\div 1024$ (to KB) $\div 1024$ (to MB)

Week 7 — Introduction to Networks

Syllabus: 4.1 – 4.2 Network Types and Hardware

Keywords: LAN 局域网 (júyù wǎng), WAN 广域网 (guǎngyù wǎng), WLAN 无线局域网 (wúxiàn júyù wǎng), PAN 个人局域网 (gèrén júyù wǎng), router 路由器 (lùróuqì), switch 交换机 (jiāohuànjī), hub 集线器 (jíxiànqì), NIC 网卡 (wǎngkǎ), modem 调制解调器 (tiáozhì jiětiáo qì)

Activity 1 (15 min): Network diagram drawing — Draw the school network. Label LAN, WAN, router, switch, server.

Activity 2 (15 min): Hardware matching — Match device to function. What's the difference between hub and switch?

Activity 3 (10 min): "Is the Internet a WAN?" Discussion — Yes! The largest WAN in the world.

Week 8 — Network Topologies

Syllabus: 4.3 Network Topologies

Keywords: bus topology 总线拓扑 (zǒngxiàn tuòpǔ), star topology 星型拓扑 (xīngxíng tuòpǔ), mesh topology 网状拓扑 (wǎngzhuàng tuòpǔ), node 节点 (jiédiǎn), backbone 骨干 (gǔgàn), terminator 终结器 (zhōngjié qì)

Activity 1 (15 min): Physical topology models — Use string and cups to build bus, star, and mesh models.

Activity 2 (15 min): Pros and cons table — For each topology: ease of setup, cost, fault tolerance, cable length.

Activity 3 (10 min): "What happens if...?" — Cable breaks in bus? Switch fails in star? One node fails in mesh?

Exam Tip: Star topology is the most common in modern networks. Always mention: central device (switch/hub), easy fault detection, but single point of failure if central device fails.

Week 9 — Network Protocols & Internet

Syllabus: 4.4 – 4.5 Protocols, IP Addresses, and the Internet

Keywords: protocol 协议 (xiéyì), TCP/IP 传输控制协议/网际协议 (chuánshū kòngzhì xiéyì/wǎngjì xiéyì), HTTP/HTTPS 超文本传输协议 (chāo wénběn chuánshū xiéyì), FTP 文件传输协议 (wénjiàn chuánshū xiéyì), IP address IP地址 (IP dìzhǐ), DNS 域名系统 (yùnmíng xìtǒng), packet 数据包 (shùjù bāo), bandwidth 带宽 (dàikuān), latency 延迟 (yánchí)

Activity 1 (15 min): "How the Internet Works" video + diagram labeling — ISP, DNS server, web server, client.

Activity 2 (15 min): Packet simulation — Students pass "packets" (paper slips) across the room. What if one is lost? (TCP retransmission!)

Activity 3 (10 min): IP address format — IPv4 vs. IPv6. Why did we run out of IPv4 addresses?

Exam Tip: For "explain how a webpage is loaded" questions, use this sequence: 1) User types URL, 2) DNS converts to IP, 3) Request sent via ISP, 4) Web server sends data in packets, 5) Browser assembles and displays page.

Past Paper Practice — Weeks 1–9 Topics

May/June 2023 Paper 1, Q1:

"State two differences between a mainframe computer and a personal computer." [2 marks]

Model answer: Mainframes support hundreds/thousands of simultaneous users; PCs support one user. Mainframes have greater processing power and storage capacity. Mainframes are more expensive.

May/June 2022 Paper 1, Q3:

"Describe the purpose of a router in a network." [2 marks]

Model answer: A router directs data packets between networks. It determines the best path for data to travel and can connect a LAN to the Internet (WAN).

Oct/Nov 2021 Paper 1, Q5:

"Explain why a solid-state drive (SSD) is more suitable than a hard disk drive (HDD) for use in a laptop." [3 marks]

Model answer: SSDs are more portable because they have no moving parts, making them more durable/shock-resistant. SSDs use less power, extending battery life. SSDs have faster read/write speeds, improving performance.